UQ School of Mechanical and Mining Engineering,
Summer Research Scholarship Report

University of Queensland

Plasmadynamics Modelling, a Two-Temperature Framework

Low Temperature Air Plasma In Eilmer Version 4.1.0

Author:
Thomas Condon

Student Number:
s4881975

February 24, 2026

Contents

List of Figures **ii**

1 Introduction **1**
1.1 Description 1

2 Methodology **2**
2.1 Assumptions 2
2.2 Two-Temperature Energy Mechanism 3
2.3 Energy Exchange Mechanisms 5
2.4 Kinetics Reaction Rate Mechanisms 6
2.5 Numerical Methods 6

3 Results **8**
3.1 Relaxation Test Case 8
3.2 Energy Investigation Test Case 10
3.3 Sensitivity Investigation Test Case 13

4 Conclusions **16**

5 Future Work **17**

References **19**

List of Figures

Figure 3.1 Relaxation of 2T air. Upper panel: Transrotational temperature T (red solid and vibroelectronic temperature T_{ve} (blue dashed) converge asymptotically to a common equilibrium. Lower panel: Neutral species mass fractions evolve from frozen air composition toward a dissociated/partial recombined mixture as equilibrium is approached 8

Figure 3.2 Validated relaxation test case for another model in the gdtk source code, following the same setup as Figure 3.1 9

Figure 3.3 Temperature evolution for all four electric field cases. The transrotational temperature T (red solid) remains near $500\,\mathrm{K}$ and slightly decreases in cases with higher electric field, as the electron heating inflates the vibroelectronic temperature T_{ve} (blue dashed). This vibroelectronic temperature rises continuously and monotonically, with the rate of increase scaling strongly with the electric field strength 10

Figure 3.4 Neutral species mass fraction evolution for all four electric field cases. N_2 (solid blue) and O_2 (solid red) remain dominant throughout the $25\,\mu\mathrm{s}$ window. N (dashed orange) and O (dashed pink) rise gradually, whilst NO (dotted grey) grows steadily through the exchange reaction rate mechanism 11

Figure 3.5 Charged species mass fraction evolution for all four electric field cases on a logarithmic scale. Free electrons e^- (solid black), N_2^+ (dashed blue), O_2^+ (dashed red), O_2^- (dot-dash green), NO^+ (dot-dash grey), N^+ (dotted yellow), and O^+ (dotted pink) are depicted 12

Figure 3.6 Mass fractions of e^- (solid purple) and O_2^- (dashed green) as a function of electric field after $25\,\mu\mathrm{s}$ for 4 temperature ratios, with an electric field step size of $1000\,\mathrm{V\,m^{-1}}$ 13

Figure 3.7 Mass fractions of e^- (solid purple) and O_2^- (dashed green) as a function of electric field after $25\,\mu\mathrm{s}$ for 4 temperature ratios, with an electric field step size of $100\,\mathrm{V\,m^{-1}}$ 13

Introduction

Plasmadynamics is one of the most sought after forms of research due to its wide breadth in application. As a Summer Research Scholar project, the Centre for Hypersonics (CFH) and School of Mechanical and Mining Engineering (SoMME) are investigating low temperature, non-thermal air plasmas for air-breathing plasma thrusters in high-speed propulsion and cold plasma jets for biomedical applications.
The coined term ABLE (Air Breathing Lorentz Engine) is of particular interest, where a theoretical hypersonic vehicle may ram into air during its flight and direct it through a small channel which has a highly powerful electron beam which smashes additional electrons into the air species moving above Mach 5. These air species are then exposed to an incredibly large electric field and magnetic field, which in turn create an induced current and then induced magnetic field, essentially forming a recurring Lorentz force which acts as a source of momentum for the air.
The main question to be answered, which PhD candidate Roshan John Kurian and CFH Co-Director Vincent Wheatley are proposing, is whether these air breathing plasma thrusters are capable of a net thrust production at these hypersonic speeds and how could this complex plasmadynamics case best be modelled to reflect reality?

Description

The problem at hand is to develop an Eilmer model which can incorporate the different dynamic behaviour of non-thermal air plasmas. This includes incorporating reaction rates defined in [8],[7],[6] into an Eilmer kinetics file, which is radically different due to its use of negative ions, enhanced rates from the reduced electric fields, and non-Arrhenius rates. This radically different method has been proved to be far more accurate for the low temperature plasmas that the CFH is interested in.
Hence, a standalone hard-coded air gas model with chemical and energy exchange mechanisms affiliated with low temperatures needs to first be made without detrimentally impacting the Eilmer source code, then tested and validated with unit cases, and lastly implemented into the Eilmer source code. The model presented in this report was made in a version of Eilmer Version 4.1.0 as opposed to the recent release of Eilmer Version 5.0.0.
The end goal of this low temperature air plasma model is to use it in 2D simulations of the ABLE to evaluate its impact on the initial modelled performance. With this evaluation, the CFH can better inform their experimental efforts on the development of low temperature, non-thermal plasmas.

Methodology

To develop a numerical model of the low temperature air plasma expected to be encountered in the ABLE, the base build and structure of Eilmer is utilised with a Two-Temperature (2T) energy mechanism. This mechanism separates the internal energy in thermochemical non-equilibrium hypersonic flow into a translational-rotational (commonly known as transrotational) energy and temperature u, T and a vibrational-electronic (commonly known as vibroelectronic) energy and temperature u_{ve}, T_{ve}. This model is advantageous for the problem at hand as it captures the chemistry and vibrational coupling which is absolutely essential to accurately model the thermal behaviour of the plasma, which is evidently one of the most critical points of the plasma.
With this mechanism and intricate reaction rates, it is important to recognise that we will require some valuable assumptions with an in depth and accurate governing Ordinary Differential Equation (ODE) scheme.

Assumptions

The numerical model and D-language program will make use of the following assumptions and design decisions, which define the scope and numerical accuracy of the model:

- The air is modelled as a 12 species mixture including negative ions present in low temperature air plasma. The 12 species are $e^-, O, O^+, O_2, O_2^+, O_2^-, N, N^+, N_2, N_2^+, NO, NO^+$ and all of their molar masses are sourced from PubChem [5] with ion molar masses calculated by finding the difference between the parent molecule and the added or removed electron molar mass

- the number density of NO^+ is not an independent variable in the ODE state vector, instead it is reconstructed every time step with the assumption of quasi-charge neutrality. This allows us to say that $n_{NO+} = n_{O+} + n_{O2+} - n_{O2-} + n_{N+} + n_{N2+}$, which assists in preventing an accumulation in floating point error

- the electron temperature T_e does not change depending on its own energy equation in the gas model. Instead, the electron temperature is calculated at each sub-step from the reduced effective electric field E^* using a nine coefficient polynomial expression (table 3 in [8]) and Horner's method to save computational power. As described in [8], when E^* is incredibly small ($<10^{-25}\,\mathrm{Vm}^2$) then T_e defaults to the gas temperature T, which improves the ability for the model to recover thermal equilibrium

- the viscosity and thermal conductivity of the low temperature air plasma is computed using the Sutherland Law for air, which ignores the ionisation fraction and is valid from the static pressure range of $0.01 - 100$ atm and a temperature range of $0\,\mathrm{K}$ to $3000\,\mathrm{K}$. Hence, it is an acceptable simplification and assumption for non-thermal low temperature air plasma in hypersonic flight, as these conditions are common and the plasma's transport properties should remain predominantly the same as the non-plasma neutral gas

- reactions are only evaluated when $T > T_{min}$, where T_{min} is the minimum temperature required for the gas to react and it can be set in the supplied Lua file

- vibroelectronic energy u_{ve} in the kinetics model is calculated as the difference between the total energy and the transrotational energy. This assumption is held true due to the thermochemical reactor being treated as an isolated reactor

- the assumption of an isolated reactor is further used in the calculation of the number density of the whole plasma itself N, which is necessary to determine the reduced effective electric field E^*

Two-Temperature Energy Mechanism

Translational and Rotational Energy and Temperature

The basis of the low temperature air plasma model is the 2T formulation which is typically used for high-enthalpy hypersonic flows found in some shock tunnel experiments (hence the groundwork for 2T models is already developed into Eilmer's Gas Dynamics Toolkit (gdtk)), with the goal of informing shock tunnel experiments of non-thermal plasma, this energy mechanism was adapted. The 2T formulation partitions the thermodynamic state as stated in the assumptions. The transrotational internal energy accounts for the translational kinetic energy of the heavy particles and their fully excited rotational degrees of freedom. The energy does not take into account the free electrons in the flow, as they contribue zero transrotational energy since their translational mode is captured in the vibroelectronic energy mode u_{ve}. For each heavy species s, the transrotational thermal energy follows the equation below which is based on the definitions found in [2]:

$$u_{tr} = \sum_s Y_s \cdot \frac{5 + 2\zeta_s}{2} \cdot (h_{tr,s} - R_s \cdot T) \tag{2.1}$$

Where, Y_s is the mass fraction of every heavy species, $\zeta_s = 0$ for atoms and $\zeta_s = 1$ for molecules (to align with the definition of the equipartition theorem in statistical thermodynamics [1]), $R_s = \frac{R_u}{M_s}$ is the specific gas constant for each heavy species, and T_{tr} is the heavy gas transrotational temperature. Also, $h_{tr,s}$ is the enthalpy of the heavy species in reference to its formation enthalpy and the corresponding temperature. For this case, the reference temperature was chosen to be 298.15 K as that is the temperature at which the gdtk species' database had formation enthalpies (ΔH_f°) for each atom or molecule in the 12 species gas model in use. These formation enthalpies were converted to $\mathrm{J\,kg^{-1}}$, and this transrotational enthalpy can be defined as:

$$h_{tr,s} = cp_{tr,s} \cdot (T - T_{ref}) + \Delta H_{f,s}^\circ \tag{2.2}$$

With the specific heat at constant pressure calculated as:

$$cp_{tr,s} = \frac{5}{2} \cdot R_s + R_m \tag{2.3}$$

Where the additional R_m term accounts for the two active rotational degrees of freedom of diatomic molecules m.

The transrotational temperature T of the heavy species can then be recovered analytically from the transrotational energy u by direct inversion, since the transrotational specific heats are constant (fully excited at all temperatures), making the temperature update a single closed-form expression rather than an iterative solution, as specified below:

$$T = \frac{u + \sum_s Y_s \cdot (cp_{tr,s} \cdot T_{ref} - \Delta H_{f,s}^\circ)}{\sum_s Y_s \cdot (cp_{tr,s} - R_s)} \tag{2.4}$$

The equation of state for the low temperature air plasma mixture requires caution since the electrons present obey a partial pressure governed by T_{ve} instead of T. The pressure is thus calculated as:

$$P = \rho \cdot ((R_{heavy} \cdot T) + (Y_e \cdot R_e \cdot T_{ve})) \tag{2.5}$$

Where R_{heavy} is the mass weighted gas constant of the heavy species, and the second term in the product of the mixture density ρ is all referring to the electron species, since it is denoted with subscript e. This formula is explicitly taken from both [8] and other gas models in the gdtk, it also correctly accounts for the Dalton's law contributions of each temperature mode to the total pressure.

Vibrational and Free Electron Energy and Temperature

The standard approach to calculate the vibroelectronic energy u_{ve} and its associated specific heat $cv_{v,ve}$ is to use NASA's Chemical Equilibrium with Applications (CEA) polynomial curve fits for thermodynamic properties. However, the CEA polynomials inherently are derived with the assumption that the mixture is at thermal equilibrium (i.e. $T = T_{ve}$).

In the non-thermal low temperature air plasma environment which can be expected with the ABLE configuration, the electron temperature and hence vibroelectronic temperature diverges heavily from the heavy particle gas temperature T by potentially thousands of Kelvin. Evaluating the proposed CEA polynomials at a T many times larger than T_{ve}, whilst the actual vibroelectronic temperature is low (and vice versa), would produce results incredibly thermodynamically inconsistent.

Therefore, the model developed deliberately does not use the pre-existing D-language gdtk architecture for CEA polynomials for the vibroelectronic mode entirely. Instead, the vibrational energy of the heaviest diatomic molecules m (O_2, N_2, NO) is computed analytically using the Vibrational Harmonic Oscillator (VHO) model used by sources like [4] and [9].

One thing that can be overlooked is the maintenance of thermodynamic consistency, by ensuring that the vibroelectronic energy is exactly 0 at the reference temperature T_{ref} =298.15 K, then the net contribution of the stored vibrational energy is 0 and it prevents accidental offsets in the total energy allowed for the modes to occupy which would occur if $u_{v,ve}$ was referenced to some other point inconsistently. This is accomplished simply by deducting the vibrational energy $u_{v,ve}$ by the reference vibrational energy at T_{ref}, u_{v,ve_0}, which is calculated in the gas model constructor. Using this model and accounting for the reference vibrational energy, the vibrational energy of the low temperature air plasma is:

$$u_{v,ve} = \sum_m Y_m \cdot \left(\frac{R_m \cdot \Theta_{v,m}}{\exp \frac{\Theta_{v,m}}{T_{ve}} - 1.0} - u_{ve_0}\right) \tag{2.6}$$

Here, $\Theta_{v,m}$ is the characteristic vibrational temperature of the heaviest diatomic molecules, set as 3353 K for N_2 [8], 2719 K for NO [9], and 2240 K for O_2 (taken from the Park equations in [10]). Then, the corresponding vibrational specific heats at constant volume follows the derivative of the VHO energy with respect to T_{ve}:

$$cv_{v,ve} = \frac{R_m \cdot (\frac{\Theta_{v,m}}{T_{ve}})^2 \cdot \exp \frac{\Theta_{v,m}}{T_{ve}}}{(\exp \frac{\Theta_{v,m}}{T_{ve}} - 1)^2} \tag{2.7}$$

Furthermore, the electron contribution to the vibroelectronic mode is computed as the translational kinetic energy of the free electrons, referenced to T_{ref} and including the electron formation enthalpy. Thus, the total vibroelectronic energy of the mixture is:

$$u_{ve} = Y_e \cdot \left(\frac{3}{2} \cdot (T_{ve} - T_{ref}) + \Delta H^{\circ}_{f_e} - R_e \cdot T_{ref}\right) + u_{v,ve} \tag{2.8}$$

Given this vibroelectronic energy u_{ve}, the vibroelectronic temperature T_{ve} can be recovered iteratively by solving the above equation for T_{ve} using a fixed iteration (Newton-Raphson up to 20 iterations with a 10^{-6} K tolerance) with $cv_{v,ve}$ as the derivative. The T_{ve} input into the functions is the initial guess, which helps with a more rapid convergence as the vibroelectronic temperature should not be changing in the orders of magnitudes within one timestep on scales of nanoseconds.

Energy Exchange Mechanisms

A critical component of the 2T low temperature air plasma model is resolving the energy transfer between the electrons in the vibroelectronic mode and the heavier particles in the transrotational mode. To better model this, two distinct physical mechanisms are used.

Elastic Energy Exchange

The elastic collisional energy transfer between free moving electrons and heavy particles is modelled using the Appleton-Bray formulation, guided by existing architecture built in the gdtk models, [4], and [3]. The elastic energy exchange rate in W/m^3 represents the net power transferred from the electron pool of energy inside of the vibroelectronic mode into the heavy transrotational mode.

To compute this energy exchange, the collision cross-sectional areas $Q_{e,k}$ for electron-neutral collisions are computed as polynomial functions of the electron temperature T_e, following [3] curve fits for O, O_2, N, N_2, NO, with k denoting these molecules.

Then, the thermal velocity of the electrons is computed from the Maxwell-Boltzmann distribution from statistical mechanics (as opposed to treating the electron velocity as the same as the bulk electron drift velocity, which is just equivalent to the total gas velocity). This distinction is of high importance as it led to the corrected computation of the collision rates with the thermal velocity, whereas the bulk electron velocity molds the directed motion of the gas with the bulk thermal motion of the gas. This average thermal electron velocity is taken from [1], defined as:

$$v_{e,th} = \sqrt{\frac{8 \cdot k_B \cdot T_e}{\pi \cdot m_e}} \tag{2.9}$$

With k_B equivalent to the Boltzmann constant, and m_e is the mass of a single electron. This then implies that the electron-neutral collision frequencies (per unit volume) are then:

$$nu_{e,k} = \frac{4}{3} \cdot Q_{e,k} \cdot n_k \cdot v_{e,th} \tag{2.10}$$

For the electron-ion collisions, the Coulomb logarithm lambda is computed from the Debeye length L_e and the electron number density, then the combined ion collision frequency $nu_{e,i}$ is computed analogously. Therefore, the total elastic energy exchange rate is then:

$$Q_e = 3 \cdot n_e \cdot m_e \cdot k_B \cdot (T_e - T) \cdot (\sum_k \frac{nu_{e,k}}{m_k} + \frac{nu_{e,i}}{m_{i,avg}}) \tag{2.11}$$

This expression ensures that when $T_e > T$, energy flows from the electron energy pool into the heavy transrotational mode (i.e. heating the gaseous mixture) and vice versa. This elastic energy rate is treated as an external source of energy to the typically ODE state vectors in kinetics models, since the vibroelectronic temperature T_{ve} is maintained by the reduced effective electric field E^* rather than evolving freely under adiabtic isolated reactor conditions.

Vibrational-Translational Relaxation

In addition to the electron-heavy elastic energy exchange rate, energy transfer between the vibrational mode of diatomic molecules and the translational mode of the gas is modelled via the Landau-Teller relaxation formulation. For each aforementioned diatomic species m (O_2, N_2, NO), the V-T relaxation rate can be computed as:

$$\frac{dQ_{VT}}{dt}_m = \frac{u^*_{v,ve_m}(T) - u_{v,ve_m}(T_{ve})}{\tau_{VT_m}} \tag{2.12}$$

Where $u^*_{v,ve_m}(T)$ is the equilibrium vibrational energy evaluated at the transrotational temperature T (calculated similarly to equation 2.6, but without the reference value) and τ_{VT_m}

is the vibrational-translational relaxation time. This relaxation time is computed using the Millikan-White formulation utilising Gnoffo A_{MW} parameters [3], with a Park high temperature correction to prevent unrealistic relaxation at very high temperatures. The net V-T energy transfer rate is accumulated into the very last state vector term, which governs the rate of change of the transrotational energy in the ODE loops.

Kinetics Reaction Rate Mechanisms

The kinetics mechanism implemented in the low temperature air plasma model is primarily based on the Parent, Shneider, and Macheret (2016) reaction rate framework [8], with support and extensions from [7] and [6]. The list of reaction rate coefficients encompass 50 different reactions which can be sorted into different classes. These are presented in extensive detail within the kinetics model itself.

A key distinguishing feature of these reaction rates compared to standard air plasma models in the gdtk and elsewhere, is the use of non-Arrhenius rate coefficients for the electron-impact ionisation reactions (1a-1e). Rather than the conventional Arrhenius form of $k = A \cdot T^n \cdot \exp \frac{-E_a}{R \cdot T}$, the Townsend Ionisation rates are expressed as exponential polynomials in the natural logarithm of the reduced effective electric field E^*. As one example, reaction 1a takes the form:

$$k1a = \exp\left(-0.0105809 \cdot (\ln E^*)^2 - 2.40411 \exp\left(-75 \cdot (\ln E^*)^{46}\right)\right)[cm^3/s] \tag{2.13}$$

Where $E^* = |E + V_e \times B|/N$ is the reduced effective electric field strength in the electron's reference frame. This value captures the combined effect of the applied electric field E in $\mathrm{V\,m^{-1}}$, the bulk electron velocity V_e in $\mathrm{m\,s^{-1}}$, and the magnetic field B in T, all normalised by the total plasma number density N in $\mathrm{m^3}$.

When there is no external electric fields, then these rate coefficients fall to zero or approximately zero, which means that this kinetics mechanism can also accurately model the absence of electron-impact ionisation in a field-free plasma at low temperatures. The electron beam ionisation rates (7a-7b) are also proportional to the normalised beam power $Q_b^* = Q_b/N$, directly linking the electron beam volumetric deposition to the ionisation source terms.

The stoichiometric ratio information for all 50 of the reactions is encoded into a static, immutable 12×50 Stoichiometric Ratio Table (SRT), which maps the production and/or consumption of each species in each reaction. The overall rate of change of the number density of some species s is then:

$$\frac{dn_s}{dt} = \sum_r SRT(r,s) \cdot k_r \cdot \Pi_j n_j^{\alpha_{j,r}} \tag{2.14}$$

Where $\alpha_{j,r}$ is the reaction order of species j in reaction r. Two-body reactions scale as $n_1 \cdot n_2$, while three-body reactions scale as $n_1 \cdot n_2 \cdot n_3$, unfortunately contributing an additional source of numerical stiffness.

Numerical Methods

The system of ODEs governing the chemical species number densities and the transrotational energy is notoriously stiff due to the enormous range of characteristic time scales present. The fastest ionisation and electron attachment reactions operate on micro and nanosecond scales, whilst recombination and V-T relaxation may take longer. To address this stiffness fairly robustly, most numerical models use an implicit Backward Euler method. However, it was recommended to simply implement two explicit ODE integration schemes (Forward Euler, and Runge Kutta 4 method) that can be user-selectable via the Lua configuration file instead.

The state vector S integrated by both methods has 12 total components. The number densities

of 11 out of 12 of the independent air model (NO^+ is the only one not included, as it is reconstructed as mentioned in the assumptions) and the transrotational internal energy take up these spots in the state vector. The derivative of S (denoted as $F(S^n)$) is computed by finding the rate of change of each variable in the state vector, such as equations 2.14 and 2.12.

Forward Euler

The Forward Euler method advances the state vector by a single explicit step of size Δt.

$$S_i^{n+1} = S_i^n + \Delta t \cdot F(S^n) \tag{2.15}$$

This method is first order accurate in time with a leading truncation error of $\mathcal{O}(\Delta t)$. This explicit scheme can be unconditionally unstable for stiff systems unless the time step is kept incredibly small relative to the reaction time scale. During initial development of the low temperature air plasma model, time steps on the order of nanoseconds were used with this method to maintain stability .

Fourth Order Runge Kutta

The RK4 method evaluates the derivative function F at four intermediate points within each sub-step, providing a fourth order accurate update. The four slope estimates are defined in the kinetics model, with the update formula coming to:

$$S_i^{n+1} = S_i^n + \frac{1}{6} \cdot (k1_i + 2 \cdot k2_i + 2 \cdot k3_i + k4_i) \tag{2.16}$$

This RK4 scheme offers substantially superior accuracy and numerical stability compared to Forward Euler for the same step size, making it the recommended integration method for use. A key detail which isn't evident in equation 2.16, is that the species number densities are floored at zero at each intermediate RK4 stage to prevent any physically impossible negative number densities. The gas state (GasState, Q) is then updated after each complete sub-step, which ensures that the thermodynamic state used to evaluate the next sub step's reaction rates is entirely consistent with that sub step's composition and energy. This eliminates a subtle instability and inaccuracy in previous versions of the low temperature air plasma model where the gas state was only updated at the end of the full time interval.

Moreover, another important detail in the energy exchange treatment of the state vector, is that the electron beam power Q_b and elastic energy exchange rate Q_e are external sources to the total energy in a 0D reactor. This implies that in the unit test cases they are explicitly added to the total energy u_{total} at each step. As noted before and in the model comments, the addition of the energy deposition from the electron beam source should be removed if the user's source term file independently accounts for this external energy.

Results

Three distinct 0D unit test cases were devised to validate the model under different physical conditions. The relaxation test case validates the ability of the model to correctly evolve a shock heated frozen air mixture toward chemical and thermal equilibrium. The energy investigation test case probes the model's behaviour in a low temperature, low pressure, and high energy scenario representative of some of the earliest simulations of the ABLE, with a comparison between 3 different electric field strengths to a base case of $0\,\mathrm{V\,m^{-1}}$. Then, the sensitivity investigation delves into the impact of different 2T ratios and electric field ranges on e^- and O_2^- mass fractions after $25\,\mu$s of chemical and thermal evolution.

Relaxation Test Case

The relaxation test case was initialised to simulate post-shock frozen air, it follows the approach of a chemistry validation test case written by Nick Gibbons. The initial conditions were $T = 12\,000\,\mathrm{K}$ (post-shock transrotational temperature), $T_{ve} = 300\,\mathrm{K}$ (frozen vibroelectronic temperature), $P = 7$ atm, and mass fractions of 76.7% N_2 and 23.3% O_2 (frozen air composition). All plasma engine parameters are not used ($E = 0\,\mathrm{V\,m^{-1}}$, $V_e = 0$ /s, $B = 0\,\mathrm{T}$, and $Q_b = 0\,\mathrm{W/m^3}$) to isolate the pure thermochemical relaxation. The system was integrated forward in time over $20\,\mu$s using the RK4 scheme, and this is what you can see below in Figure 3.1.

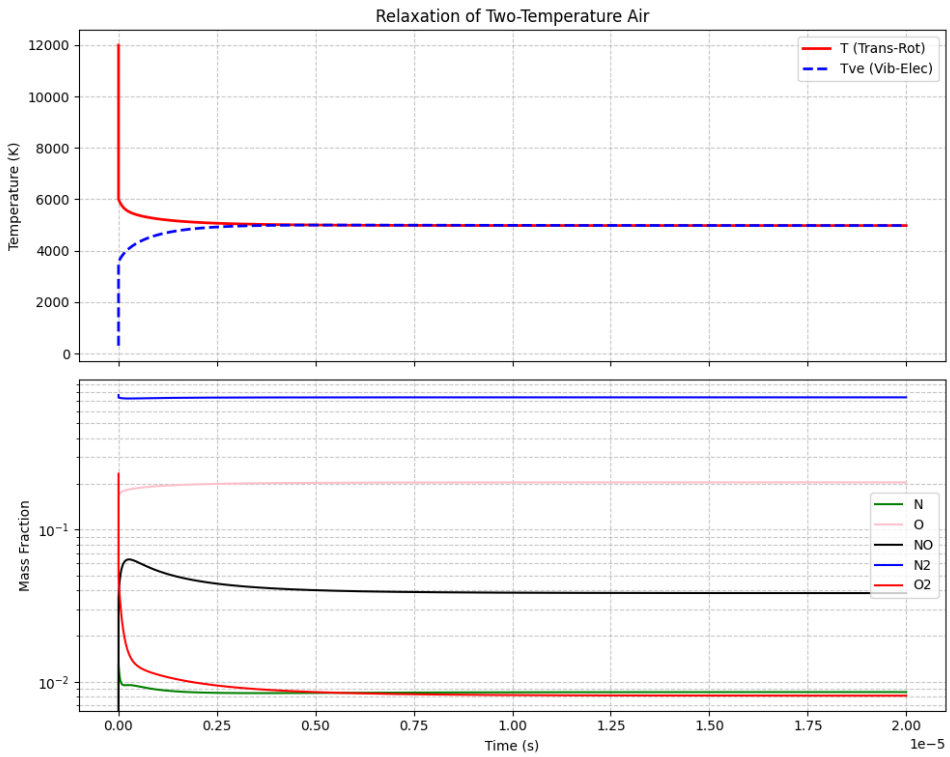

Figure 3.1: Relaxation of 2T air. Upper panel: Transrotational temperature T (red solid and vibroelectronic temperature T_{ve} (blue dashed) converge asymptotically to a common equilibrium. Lower panel: Neutral species mass fractions evolve from frozen air composition toward a dissociated/partial recombined mixture as equilibrium is approached

As demonstrated above, the transrotational temperature T drops sharply from $12\,000\,\mathrm{K}$ in a few microseconds as the rapid oxygen and nitrogen dissociation reactions (8a-8l) consume

transrotational energy to supply the activation energy (in their Arrhenius rates) required to break molecular bonds. Simultaneously, the vibroelectronic temperature T_{ve} rises from 300 K as energy is transferred into the vibrational modes through the Landau-Teller V-T relaxation mechanism. The two temperatures converge asymptotically to approximately 5000 K, demonstrating that the implementation of both the transrotational energy inversion and the vibroelectronic iteration solution are functioning correctly.
In the lower panel, it is evident that the expected dissociation and recombination dynamics of the neutral species occurs. The initial N_2 mass fraction falls steeply as nitrogen dissociation (8g-8l) dominates in the early time period of the relaxation. Atomic oxygen (O) is produced from O_2 dissociation, and nitric oxide (NO) builds up rapidly through the exchange reactions 8r-8u, before settling at its equilibrium mass fraction.
This model can be validated on its own since the temperatures converge onto each other by the end of the solution, and the oxygen mass fraction exceeds 1%. However, the original test case made by Nick Gibbons also provided an expected relaxation plot, which can be seen below in Figure 3.2.

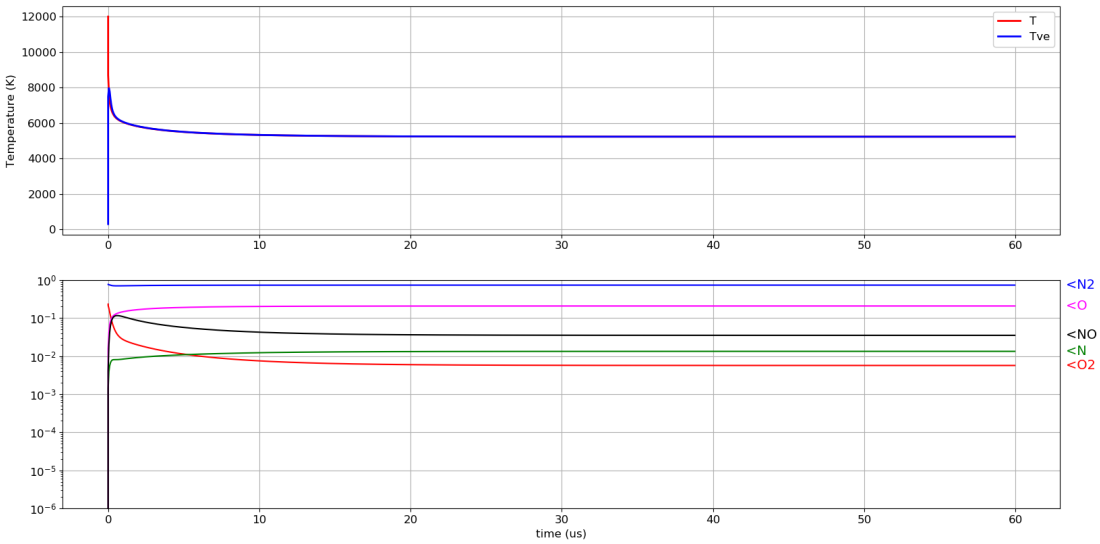

Figure 3.2: Validated relaxation test case for another model in the gdtk source code, following the same setup as Figure 3.1

The comparison demonstrates that the low temperature air plasma model behaves exactly as it should be, the only deviation from the expected result is the major jump in vibroelectronic temperature. However, this may be attributed to the ODE method used in the validated test.

Energy Investigation Test Case

The energy investigation test case was initialised with conditions representative of the ABLE's low temperature and low pressure environment. The initial conditions were $T = T_{ve} = 500\,\mathrm{K}$. $P = 2000\,\mathrm{Pa}$, with a predominantly frozen air composition (79% N_2 and 21% O_2) with a small ionisation fraction of 10^{-4} (which seeds trace amounts of O_2^+, N_2^+, and e^-). Four cases were run in parallel, each with a fixed electron beam volumetric power deposition of $Q_b = 30\,\mathrm{MW/m^3}$, a bulk electron velocity of $V_e = 2000\,\mathrm{m\,s^{-1}}$, a magnetic field of $B = 1\,\mathrm{T}$. and varying applied electric fields. Case 0 has $E = 0\,\mathrm{V\,m^{-1}}$, case 1 has $E = 15\,000\,\mathrm{V\,m^{-1}}$, case 2 has $E = 17\,500\,\mathrm{V\,m^{-1}}$, and case 3 has $E = 20\,000\,\mathrm{V\,m^{-1}}$. All cases were integrated for $25\,\mu\mathrm{s}$ using the RK4 scheme with a time step of $dt = 1\,\mathrm{ns}$.

Temperature Evolution

The temperature evolution plot for this unit test case is one of the most physically significant results, which can be seen below in Figure 3.3.

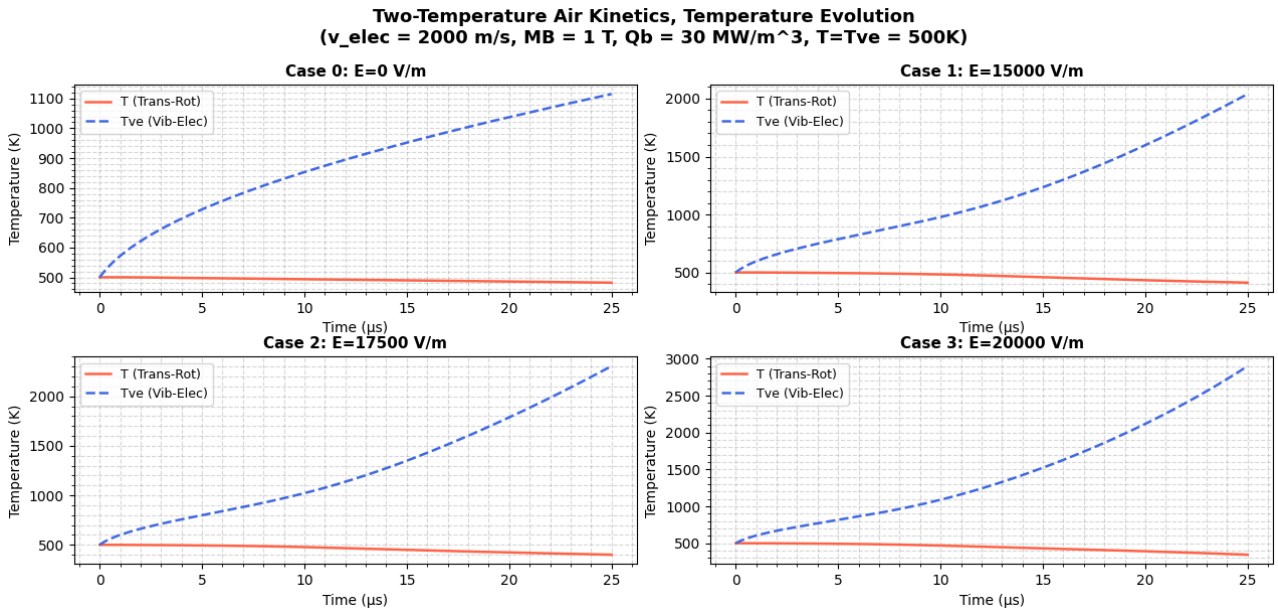

Figure 3.3: Temperature evolution for all four electric field cases. The transrotational temperature T (red solid) remains near 500 K and slightly decreases in cases with higher electric field, as the electron heating inflates the vibroelectronic temperature T_{ve} (blue dashed). This vibroelectronic temperature rises continuously and monotonically, with the rate of increase scaling strongly with the electric field strength

In case 0 (no electric field), the electron beam alone drives the vibroelectronic temperature up by approximately 1100 K in 15 μs, whilst the transrotational temperature remains near 500 K. As the applied electric field increases through cases 1 to 3, the Appleton-Bray elastic energy exchange rate grows substantially, driving progressively more energy into the vibroelectronic mode. By the third case ($E = 20\,000\,\mathrm{V\,m^{-1}}$), T_{ve} approaches 2900 K whilst T slightly decreases below its initial temperature, demonstrating that the energy deposition into the vibroelectronic mode is defining characteristic non-thermal plasma operation. This can confirm that the Appleton-Bray elastic energy exchange is correctly directing energy into T_{ve}, and it also demonstrates that this initial design decision to replace the CEA polynomials with the VHO model is functioning correctly at these highly dissimilar temperatures.

Neutral Species Kinetics

A plot of the kinetics of the neutral species (seen below in Figure 3.4) is valuable as it can corroborate with a charged species kinetics plot to demonstrate how an electric field can influence the reaction rates of air with a direct comparison to a case without an electric field.

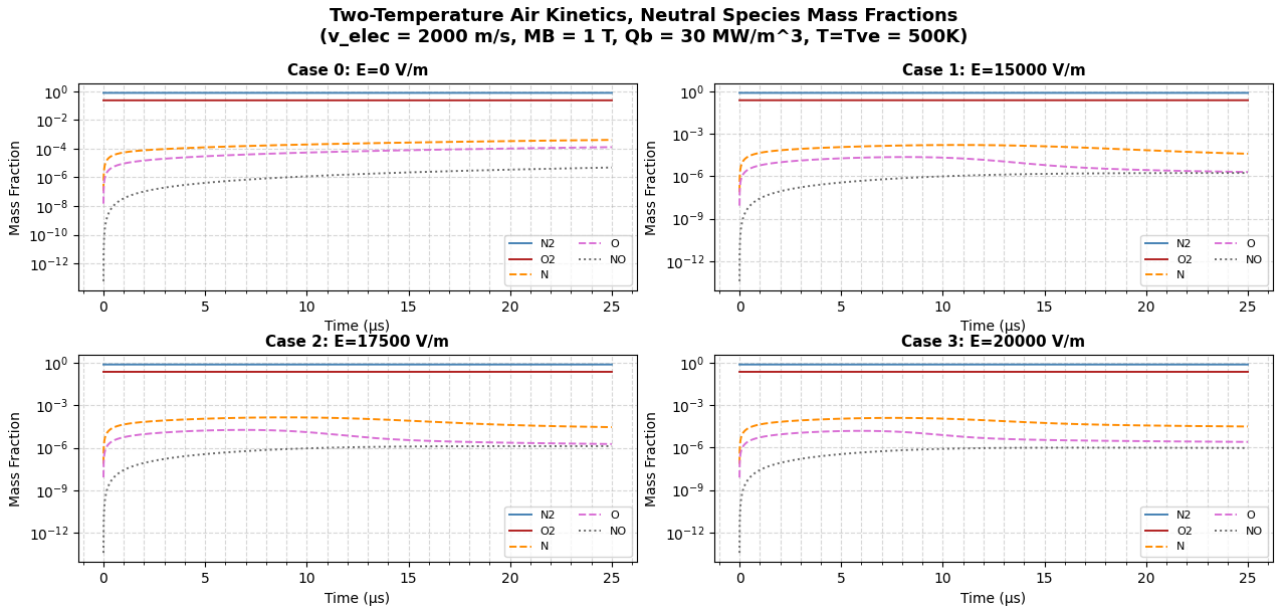

Figure 3.4: Neutral species mass fraction evolution for all four electric field cases. N_2 (solid blue) and O_2 (solid red) remain dominant throughout the 25 μs window. N (dashed orange) and O (dashed pink) rise gradually, whilst NO (dotted grey) grows steadily through the exchange reaction rate mechanism

Figure 3.4 illustrates that the bulk air composition (N_2 and O_2) remain essentially constant, reflecting the low dissociation rates of those species at only 500 K transrotational temperature. The oxygen and nitrogen dissociation reactions (8a-8l) are exponentially suppressed at these temperatures, and as such the neutral chemistry is dominated by the lower activation energy exchange reactions. NO builds steadily through the nitrogen and oxygen exchange reaction pathways, with the rate of this increasing also building up over the different cases. In case 0, the neutrals chemistry is essentially passive on this timescale, whilst in cases 1 to 3 the slightly elevated T_{ve} drives marginally higher electron beam ionisation, in turn slightly increasing the number density of atoms through associative ionisation backward reactions. The qualitative similarity across all four cases in the neutral species confirm that the dominant plasma kinetics at these conditions is entirely in the charged species, which can be observed in the following section.

Charged Species Kinetics

The charged species kinetics plot shown in Figure 3.5 below provides a rigorous validation of the implementation of the kinetic reaction rate mechanism for the low temperature air plasma model.

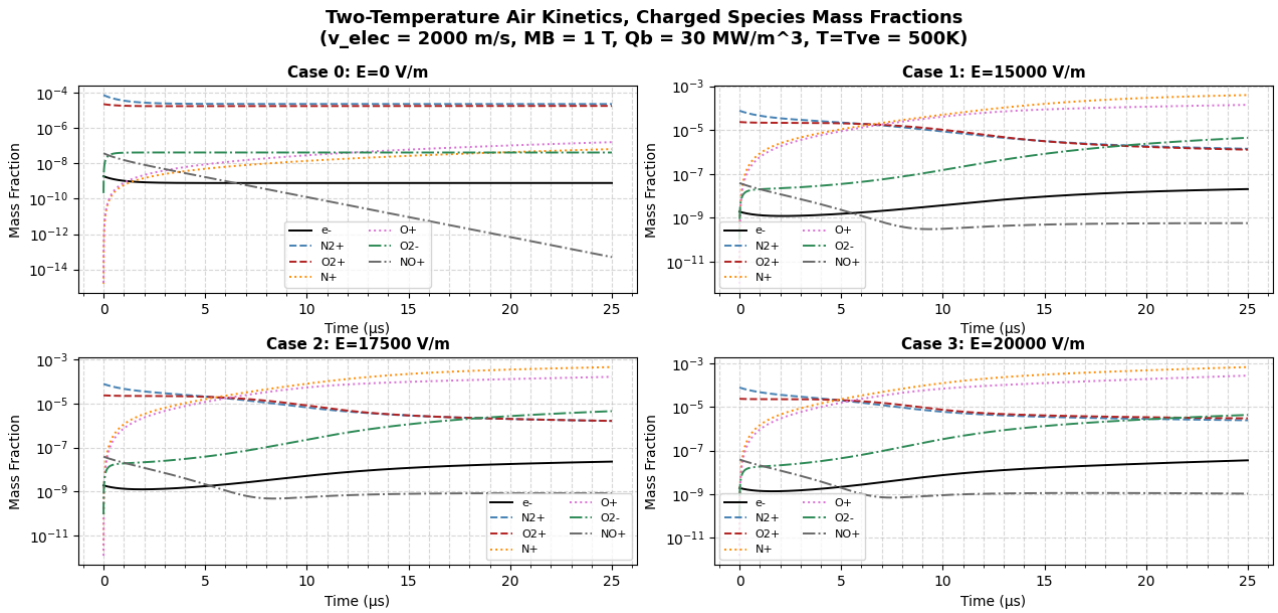

Figure 3.5: Charged species mass fraction evolution for all four electric field cases on a logarithmic scale. Free electrons e^- (solid black), N_2^+ (dashed blue), O_2^+ (dashed red), O_2^- (dot-dash green), NO^+ (dot-dash grey), N^+ (dotted yellow), and O^+ (dotted pink) are depicted

In case 0, the electron beam (7a and 7b) sustains what appears to be a steady-state ion population dominated by N_2^+ and O_2^+, with NO^+ gradually building through the associative ionisation (reaction 10). The free electron mass fraction stabilises as the beam ionisation is balanced by three body and electron-ion recombination. Then it is clear that the trace O_2^- population arises from electron attachment (5a and 5b) and is partially balanced by detachment (reaction 6).

As the applied electric field increases from case 0 to case 3, the reduced electric field E^* grows, driving the Townsend ionisation rate coefficients (1a-1c) dramatically upward. The result is a very pronounced growth in N^+ and O^+ via electron impact ionisation of the dissociated atoms, as well as a stronger NO^+ build up. Moreover, the free electron mass fraction grows by several orders of magnitude between the base case and case 3, which confirms that the reduced effective electric field is the primary driver of plasma ionisation in the ABLE channel. The successful tracking of O_2^- as a trace species across all cases without numerical failure demonstrates the stability of the RK4 scheme and the robustness of the floored number density applied to the state vector at each intermediate stage. It also successfully demonstrates the major increase in the negative ion when electric fields are applied. However, it does not seem to increase very much depending on the electric field magnitude.

Sensitivity Investigation Test Case

This test was designed to probe the competing roles of the 2T ratio T/T_{ve} and the applied electric field strength on the mass fractions of the two negatively charged species in the low temperature air plasma model (e^- and O_2^-). Rather than evolving a single state in time, this test sweeps through the electric field from $0\,\mathrm{V\,m^{-1}}$ to $25\,000\,\mathrm{V\,m^{-1}}$ and records the negative ion mass fractions after $25\,\mu\mathrm{s}$ at each field strength, for four distinct initial 2T ratios, and a time step of $dt = 1\,\mathrm{ns}$. The shared electromagnetic parameters are the same of that in Section 3.2. The results below exhibit the same test but on different electric field step scales, Figure 3.6 is 25 total steps (i.e. $1000\,\mathrm{V\,m^{-1}}$ step size) and Figure 3.7 is 250 total steps (i.e. $100\,\mathrm{V\,m^{-1}}$ step size).

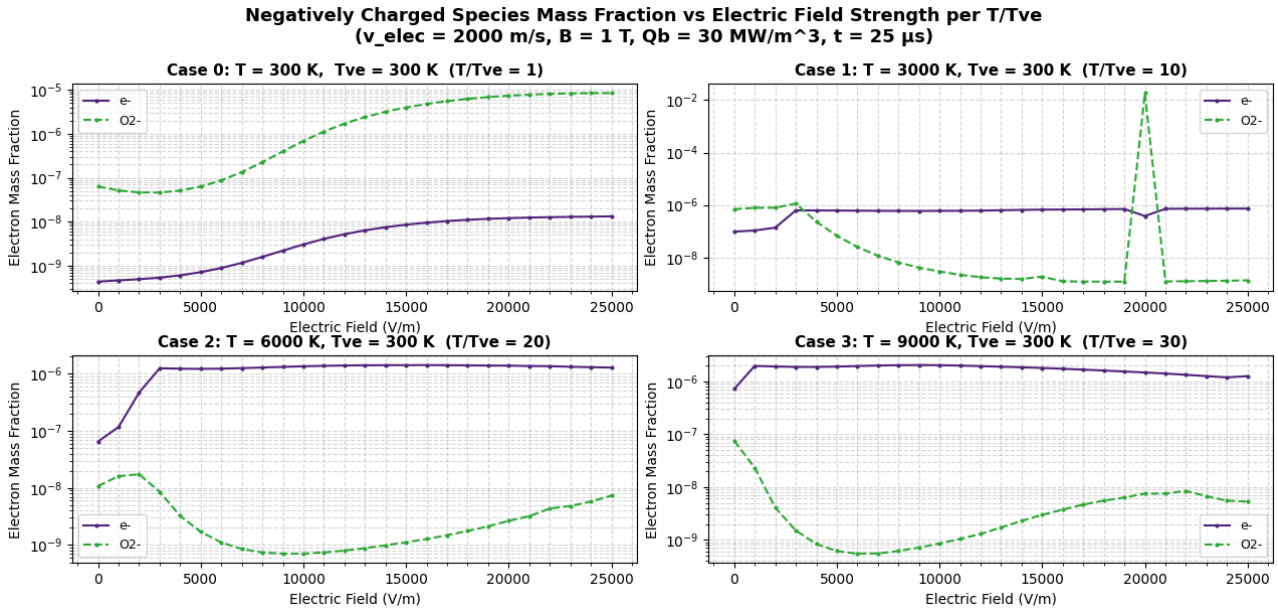

Figure 3.6: Mass fractions of e^- (solid purple) and O_2^- (dashed green) as a function of electric field after $25\,\mu\mathrm{s}$ for 4 temperature ratios, with an electric field step size of $1000\,\mathrm{V\,m^{-1}}$

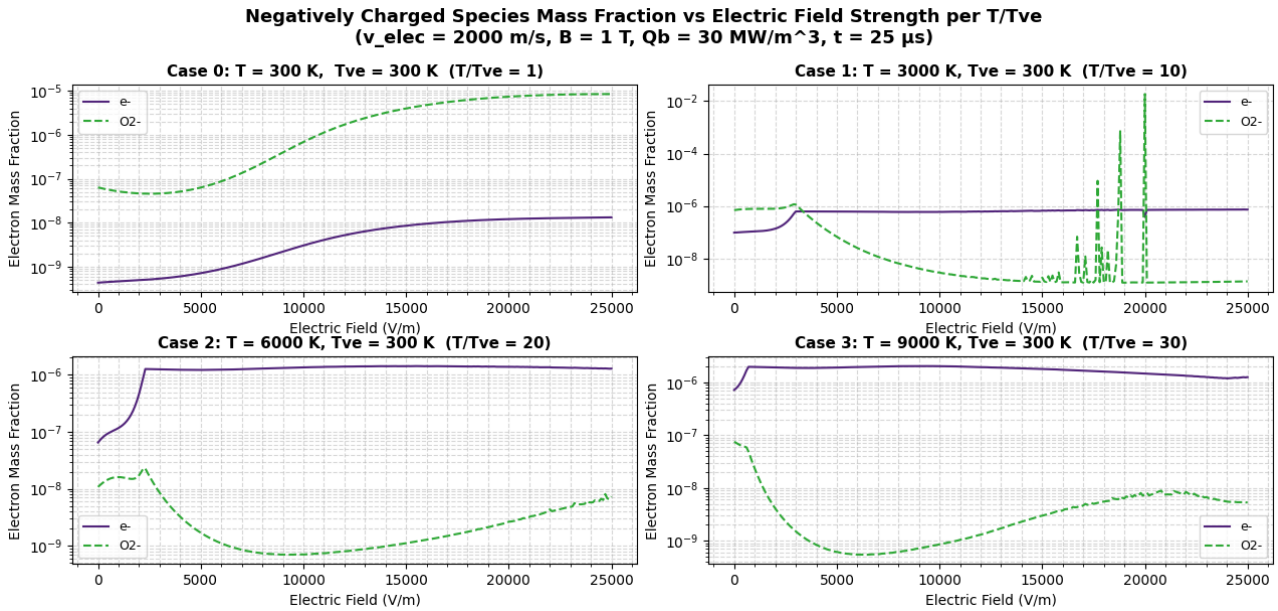

Figure 3.7: Mass fractions of e^- (solid purple) and O_2^- (dashed green) as a function of electric field after $25\,\mu\mathrm{s}$ for 4 temperature ratios, with an electric field step size of $100\,\mathrm{V\,m^{-1}}$

The four cases fix the vibroelectronic temperature $T_{ve} = 300\,\mathrm{K}$ whilst progressively increasing the transrotational temperature. Case 0 has a ratio of $T/T_{ve} = 1$ (i.e. $T = 300\,\mathrm{K}$), case 1 has a ratio of $T/T_{ve} = 10$ (i.e. $T = 3000\,\mathrm{K}$), case 2 has a ratio of $T/T_{ve} = 20$ (i.e. $T = 6000\,\mathrm{K}$), and case 3 has a ratio of $T/T_{ve} = 30$ (i.e. $T = 9000\,\mathrm{K}$). This parametric structure allows for a direct assessment of how increasing degrees of 2T non-equilibrium alter the attachment and ionisation balance across the field range relevant to the ABLE operating regime.
In case 0, where the 2T regime is at thermal equilibrium, both e^- and O_2^- increase monotonically with the applied electric field. The O_2^- mass fraction rises from approximately 10^{-7} at zero electric field to 10^{-5} at $25\,000\,\mathrm{V\,m^{-1}}$, consistently dominating over the free electron fraction, which grows more slowly from 10^{-9} to approximately 10^{-8} across the same range. This dominance of the negative molecular ion over free electrons at low temperature and low field is physically consistent, as the electron attachment reactions (5a and 5b) proceed efficiently at low T_{ve}, as the low electron temperature T_e (coupled with the E^* polynomial) keeps the detachment rate coefficient of reaction 6 suppressed. At equilibrium temperatures, the plasma is attachment dominated (and experiences recombination), so the O_2^- population grows in tandem with the increasing ionisation source from the Townsend ionisation rates as the electric field increases.

Case 1 reveals considerably more physically important behaviour. At low electric field strengths ($< 3000\,\mathrm{V\,m^{-1}}$), O_2^- reaches a pronounced peak at approximately 10^{-6}, a few orders of magnitude higher than in case 0. This is a consequence of the elevated transrotational temperature of 3000 K substantially increasing the rate of neutral chemistry through the exchange and dissociation reactions (8a-8u), which in turn raises the number density of atomic oxygen and nitrogen available to participate in three body attachment reactions. However, as the electric field increases beyond this peak, the O_2^- mass fraction collapses sharply by several orders of magnitude, reaching a minimum of just above 10^{-9} in the intermediate field range. This reversal reflects the onset of efficient Townsend ionisation (1a-1e) at elevated reduced effective electric fields E^*, which greatly accelerates electron production and shifts the plasma toward an ionisation dominated mixture. The electron detachment from reaction 6 also becomes more prevalent as the electron temperature T_e rises with E^*, actively stripping electrons from the O_2^- population. The electron mass fraction in case 1 is comparatively flat across the sweep, staying around 10^{-6}, suggesting that ionisation and recombination rapidly reach a quasi-steady balance that is not strongly sensitive to the field at this temperature.

Comparing Figures 3.6 and 3.7 for case 1, the most significant difference is the appearance of the sharp numerical oscillations in the O_2^- curve in the electric field range of approximately $15\,000\,\mathrm{V\,m^{-1}}$ to $20\,000\,\mathrm{V\,m^{-1}}$. In Figure 3.6 the O_2^- curve recovers from its minimum with a single sharp spike before settling back down, whereas in Figure 3.7 the same recovery behaviour features multiple high amplitude oscillations. This region corresponds to a transition zone where the competing time scales of attachment, detachment, and Townsend ionisation become comparable alongside the parametric electric field stepping. This must be creating a highly stiff system for which the explicit RK4 scheme struggles to resolve smoothly for this particular combination of temperature ratios and electric field.
This highlights a known limitation of the explicit integration approach. Whilst the RK4 scheme with floored number densities is robust over the broad parameter space, there must exist narrow transition regimes in E^* and T/T_{ve} space where an implicit (like Backward Euler) or adaptive step iterator would be preferable. This is a clear motivation for continued work on this low temperature air plasma kinetics mechanism.

Cases 2 and 3 both exhibit more monotonically stable responses compared to case 1. In

both cases, the electron mass fraction saturates rapidly to approximately 10^{-6} at low applied electric fields and remains essentially constant across the electric fields, suggesting that at these high transrotational temperatures the collisional ionisation from the neutral chemistry reactions (8a-8q) alone is sufficient to sustain a quasi-steady electron population independent of the applied field. This is also where the test cases from Section 3.2 can be reintroduced to investigate whether the applied electric field does still influence the speed that the electron mass fraction reaches this quasi-steady point.
The O_2^- mass fraction in both cases decreases from a moderate initial value just below 10^{-7} at low applied electric fields to a minimum of approximately 10^{-9} in the intermediate electric field range as detachment accelerates with the rising T_e, and then it gradually recovers back toward 10^{-8} at the top end of the electric fields. The absence of the large low-field O_2^- 'well' in case 1 is consistent with the higher transrotational temperatures driving rapid dissociation of O_2 (8a-8f), reducing the neutral O_2 'reservoir' available for attachment. Overall, the behaviour of these cases confirms that at high T/T_{ve} ratios representative of strong post-shock non-equilibrium, the plasma is neither attachment dominated nor strongly ionisation field sensitive in the negatively charged species, but instead governed principally by the neutral dissociation chemistry at the elevated transrotational temperature.

Overall, the sensitivity investigation demonstrated that the 2T ratio could be a critical control parameter for the negative species balance in the low temperature air plasma model. The transition from attachment dominated (case 0) behaviour to dissociation suppressed attachment with Townsend ionisation drive collapse (cases 1-3) behaviour appears physically consistent with the expected plasmadynamic chemistry, and the model captures the non-trivial dependence correctly across the parameter space. Furthermore, the numerical oscillations identified in case 1 would have to be an artefact of the explicit scheme operating at the boundary of its stability region, and does not undermine the validity of the low temperature air plasma across a broader operating range.
Lastly, investigation was taken into flipping the ratios (i.e. $T = 300\,\mathrm{K}$ and $T_{ve} = 9000\,\mathrm{K}$), but these led to no conclusions as they would simply replicate similar behaviour to case 0 in Figures 3.6 and 3.7.

Conclusions

A robust, low temperature air plasma model (with a two-temperature energy mechanism) was successfully implemented into the Eilmer 4.1.0 computational framework via two custom D-language models. The gas model establishes the foundational thermodynamic behaviour, including species definitions for a 12 species plasma, the Vibrational Harmonic Oscillator formulation for diatomic specific heats, properly referenced vibroelectronic energies anchored at $T_{ref} = 298.15\,\mathrm{K}$ and a fixed iterative solver for the vibroelectronic temperature T_{ve}. By abandoning the CEA equilibrium polynomials in favour of the VHO model, the gas model correctly handles the extreme temperature disparities characteristic of non-thermal plasma without producing the thermodynamically inconsistent results that CEA extrapolation could yield.

The kinetics model implements 50 reaction rates drawn from subsequent Parent papers ([8], [7], and [6]), including non-Arrhenius Townsend ionisation rates driven by the reduced effective electric field E^*, electron beam ionisation, negative ion chemistry involving O_2^-, and the full catalogue of dissociation, recombination, and exchange reactions for air. The Appleton-Bray elastic energy exchange, computed using thermal electron velocities from the Maxwell-Boltzmann distribution, correctly channels energy from the vibroelectronic pool into the heavy translational mode based on the instantaneous temperature disparity. Landau-Teller V-T relaxation using the Millikan-White with Park high temperature correction formula completes the inter-mode energy coupling.

Zero-dimensional unit testing confirmed that the numerical integration is highly stable and physically consistent. The relaxation test case demonstrated correct two-temperature convergence from a severely non-equilibrium post-shock state, with species mass fractions evolving along physically expected pathways. The energy investigation cases demonstrated the model's ability to maintain and amplify thermal non-equilibrium under combined electron beam and electric field forcing, with charged species kinetics tracked stably down to mass fractions of order 10^{-14}. The RK4 integration scheme is clearly superior to Forward Euler in terms of accuracy for the same time step size, whilst both schemes remained reasonably stable at nanosecond time steps for stiff test cases.
However, this became more complicated for the sensitivity investigation, which was the most physically nuanced validation, revealing that the two-temperature ratio T/T_{ve} is a critical control parameter for the negative species balance in the plasma. The plasma transitions from attachment dominated to ionisation dominated behaviour as the transrotational temperature T increases, in a physically consistent manner. It was also confirmed that inverting the ratios simply replicates the equilibrium case 0 behaviour, since the kinetically active pathways are governed by T and E^* rather than T_{ve} directly. This case was also crucial as it experienced numerical instabilities in the O_2^- curve near the attachment to ionisation transition, which are artefacts of the explicit RK4 scheme as opposed to an implicit scheme like Backward Euler. Thankfully, it is only a numerical artefact and not an indication of the model validity.

In conclusion, the model provides a physically rigorous and numerically robust foundation for advancing to 2D CFD simulations of the ABLE channel, and is ready to be integrated into Eilmer's flow solver for performance evaluation.

Future Work

With the zero dimensional kinetics and energy exchange source terms validated, several natural extensions of this work are identified for future development:

- **Validate the low temperature air plasma model based on experimental data from T4**
  - The model should be validated against experimental measurements currently being obtained from the T4 reflected shock tunnel at UQ. Specifically, taking measurements of electron number densities and emission signatures from seeded plasma shots could be compared against 0D post shock relaxation predictions from the model. This would provide confidence in both the kinetics mechanism and the 2T energy mechanism as the model is progressed through to full 2D CFD simulations. However, the development of T4 experiments is still in progress, so it may be more efficient to determine an already verified unit test case of low temperature air plasma, and then use the model presented as a way to inform the development of T4 ABLE experiments
- **Investigate the interface between spatial CFD flow fields and the model's source term evaluations**
  - The current model operates as a standalone 0D reactor and must be coupled with Eilmer's flow solver as a thermochemical source term which handles the reaction rates, energy exchange, and chemistry. This requires careful implementation of the source term files and job files with the low temperature air plasma model itself, it should interact with the reactor similarly to the unit test cases (setting electric and magnetic terms to the public variables in the kinetics file)
- **Implement a Backward Euler method**
  - The numerical oscillations observed in the sensitivity investigation near the attachment to aionisation transition reveal the fundamental limitation of explicit integration for stiff ODE systems. An implicit Backward Euler scheme with fixed iteration on the full Jacobian of the state vector would unconditionally stabilise these transition regions and permit significantly larger time steps, substantially reducing computational cost for 2D simulations
- **Implement a Three-Temperature model**
  - The current 2T mechanism couples the vibrational energy of the diatomic molecules and the translational energy of the free electrons into a single vibroelectronic temperature T_{ve}. A three-temperature (3T) extension would resolve the electron temperature T_e, the vibrational temperature T_{ve}, and the transrotational temperature T independently, which is particularly important for accurately capturing the elctron-vibrational energy exchange pathways in the N_2 molecules as recommended by [9]
- **Utilise the model to inform cost functions of the flow acceleration against joule heating**
  - A parametric investigation similar to Section 3.3 could be used to investigate how the applied electric field E at representative ABLE channel conditions can be used

to map the trade off between flow acceleration and joule heating. This would allow in a clear direction for the experiments to go to achieve net thrust results without damaging experimental equipment

- **Investigate data to determine a reliable formation enthalpy of O_2^-**
  - The formation enthalpy of O_2^- may be critical for the correct vibroelectronic energy at the reference temperature. The current value of $0\,\mathrm{J\,kg^{-1}}$ appears to not detrimentally impact the low temperature air plasma model, but more research needs to be done to confirm thermodynamic consistency
- **Implement a different V-T relaxation time calculation found to be more accurate for low temperature plasmas in [8]**
  - The current implemented relaxation time is not preferred for low temperature plasmas. Instead, [8] recommends using a relaxation time formulation specifically calibrated for the expected plasma regime (seen below), which could improve the accuracy of V-T energy coupling at temperatures less than $3353\,\mathrm{K}$

$$\frac{1}{\tau_{vt}} = 7 \cdot 10^{-16} \cdot N \cdot \exp\left(-141 \cdot T^{-1/3}\right) + 5 \cdot 10^{-18} \cdot N_O \cdot \exp\left(-128 \cdot T^{-1/2}\right) \tag{5.1}$$

- **Implement an updated numerical low temperature plasma model from [9] which utilises key parameters of the field flow and incorporates electron heating and cooling**
  - [9] recommends the use of an updated model which should be implemented that derives the electron heating rate from the corresponding electron-vibrational (e-V) cooling rate via the principle of detailed balance. The proposed formulation yields a simple heating-to-cooling ratio of $\exp\left(\frac{\theta_e}{T_e} - \frac{\theta_v}{T_v}\right)$, where θ_v is the characteristic vibrational temperature of the species, which guarantees that the electron temperature converges correctly to the vibrational temperature at thermal equilibrium. This updated model accounts for inelastic electron energy losses to all neutral molecules, including O_2 and NO, which has been shown to meaningfully lower the predicted electron temperature and improve agreement with test data. The updated model also incorporates equation (6) from [9] which utilises characteristic parameters of the field to prescribe a more accurate energy exchange
- **Investigate the decision for one of the Gnoffo, 1989 collision cross sections to be a different magnitude than what is defined in [3]**
  - The oxygen polynomial collision cross-section curve fits has a second order constant magnitude defined in one of the old Eilmer ABLE job source term files as 2.0×10^{-28}. However, it is defined in [3] as being 2.0×10^{-29}, this needs to be investigated in case it impacts the energy exchange rates

References

[1] Phil Attard. "Chapter 7 - Interacting Particles". In: *Thermodynamics and Statistical Mechanics*. Ed. by Phil Attard. London: Academic Press, 2002, pp. 153–179. ISBN: 978-0-12-066321-7. DOI: https://doi.org/10.1016/B978-012066321-7/50007-6. URL: https://www.sciencedirect.com/science/article/pii/B9780120663217500076.

[2] Vincent Casseau et al. "A Two-Temperature Open-Source CFD Model for Hypersonic Reacting Flows, Part One: Zero-Dimensional Analysis". In: *Aerospace* 3.4 (2016). ISSN: 2226-4310. DOI: 10.3390/aerospace3040034. URL: https://www.mdpi.com/2226-4310/3/4/34.

[3] Peter A Gnoffo, Roop N Gupta, and Judy L Shinn. *Conservation equations and physical models for hypersonic air flows in thermal and chemical nonequilibrium*. Tech. rep. NASA-TP-2867. NASA Langley Research Center, Hampton, VA, 1989.

[4] Yusuke Imamura and Takayasu Fujino. "Numerical Simulation of Magnetohydrodynamic Flow Control in Reentry Flight with Three-Temperature Model". In: *2018 AIAA Aerospace Sciences Meeting*. AIAA, 2018.

[5] Sunghwan Kim et al. "PubChem 2025 update". In: *Nucleic Acids Research* 53.D1 (2025), pp. D1516–D1525. DOI: 10.1093/nar/gkae1059. URL: https://doi.org.

[6] B. Parent. "Electron heating and cooling in hypersonic flows". In: *Physics of Fluids* 33.4 (Apr. 2021), p. 046105. ISSN: 1070-6631. DOI: 10.1063/5.0046197. eprint: https://pubs.aip.org/aip/pof/article-pdf/doi/10.1063/5.0046197/13634016/046105_1_online.pdf. URL: https://doi.org/10.1063/5.0046197.

[7] Bernard Parent and Kyle M. Hanquist. "Plasma Sheath Modelling for Computational Aerothermodynamics and Magnetohydrodynamics". In: *International Journal of Computational Fluid Dynamics* 35.5 (2021), pp. 331–348. DOI: 10.1080/10618562.2021.1949456.

[8] Bernard Parent, Mikhail N. Shneider, and Sergey O. Macheret. "Detailed Modeling of Plasmas for Computational Aerodynamics". In: *AIAA Journal* 54.3 (2016), pp. 898–911. DOI: 10.2514/1.J054624.

[9] Felipe Martin Rodriguez Fuentes and Bernard Parent. "Vibrational-electron heating in plasma flows: A thermodynamically consistent model". In: *Physics of Fluids* 37.9 (Sept. 2025), p. 096141. ISSN: 1070-6631. DOI: 10.1063/5.0285170. eprint: https://pubs.aip.org/aip/pof/article-pdf/doi/10.1063/5.0285170/20705244/096141_1_5.0285170.pdf. URL: https://doi.org/10.1063/5.0285170.

[10] Jesse W. Streicher, Ajay Krish, and Ronald K. Hanson. "Coupled vibration-dissociation time-histories and rate measurements in shock-heated, nondilute O2 and O2–Ar mixtures from 6000 to 14000K". In: *Physics of Fluids* 33.5 (May 2021), p. 056107. ISSN: 1070-6631. DOI: 10.1063/5.0048059. eprint: https://pubs.aip.org/aip/pof/article-pdf/doi/10.1063/5.0048059/16093945/056107_1_online.pdf. URL: https://doi.org/10.1063/5.0048059.